

and scrotum requiring operation. Four patients had relapses. One patient had several very severe intestinal hemorrhages, the first following about thirty-six hours after the first vaccine inoculation. The subsequent vaccine inoculations, however, were well borne, so that it would be unfair to assume the principle of "*post hoc ergo propter hoc*." With this possible exception, no manifestations of a negative phase were observed and no other complications noted. Only in two out of the seventeen cases did the temperatures subside acutely, reaching normal forty-eight to seventy-two hours after the vaccine inoculation; in the great majority, the temperature did not go any higher, but began to show remissions, coming down to normal gradually. It would be futile to tabulate the results of the treatment in reference to any other individual symptoms, as such detailed differences should only be based on a very much greater material. On the whole, the impression was gained that the inoculated patients ran what might be termed mild courses with comparatively few complications. I am fully aware how guarded one should be about the results of any form of specific therapy in a disease like typhoid fever. Normally, all types of illness may exist. Clinicians have frequently observed that during some typhoid seasons the patients will present mild infections without any special treatment whatever. One could attribute beneficial effects to vaccine treatment with certainty only if rapid improvement or crisis in the course of the disease would set in soon after the inoculation. Such acute changes can be expected more often by treatment with a specific serum than with a vaccine, for by the former (passive immunization) antibodies are injected ready for neutralizing the poison; while in the latter instance, the antibodies must first be manufactured by the tissue cells. This ever-existing factor in vaccine therapy makes it absolutely necessary to start the treatment in typhoid fever as early in the course of the disease as possible, at a time when the reactive power of the individual is still responsive and unimpaired by the infection.

SUMMARY

In summary it may be said:

1. The treatment of typhoid fever by sensitized typhoid bacteria offers a more rational therapy from an immunologic point of view than by the ordinary non-sensitized vaccine.
2. The repeated inoculation of large doses of sensitized vaccine even in very sick patients, was not attended by any harmful effects or a distinct negative phase; the general course of the disease seemed milder and the complications less; the occurrence of a crisis in the infection took place in a small percentage of cases, the improvement usually being gradual. Perhaps more striking results will be attained by a larger number of smaller doses more often repeated.²⁰

20. The patients referred to were treated in 1911, but publication was deferred until a series could be treated by smaller and more frequently repeated doses. This was not, however, possible.

Geographic Distribution of Diseases.—In certain diseases, of which the cause or means of spread is unknown, morbidity reports show their geographic distribution and varying prevalence and the conditions under which cases occur. This information has great potential value in attempts to ascertain their causes and means of spread.—John W. Trask, Supplement No. 12, Public Health Reports, April 3, 1914.

GUIDES IN VACCINE THERAPY

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Since the opsonic index variation method proved to be practically unsuited as a guide in the general application of vaccine therapy, most users of vaccines have not attempted to discover individual indications as to dosage, but have established a general routine initial dose, to be increased by a certain amount at a certain interval, and have persistently carried out their program as long as the treatment was adhered to. For this reason one object of the present article is to gather and describe such symptoms and manifestations as there is reason to believe give clue to that which takes place in the body following vaccine administration, and which may reliably guide to individual case variation of both dose and interval.

In general there has been a tendency to overdo and abuse this form of treatment which, in many instances, when properly handled, proves of immense value; and through frequent bungling and total disregard of the principles involved there is great danger that it will be brought into disrepute with the profession at large. To aid in counteracting this I am offering the record of my experience and interpretation of results obtained by other physicians for whom I have prepared vaccines, and I feel certain that the facts noted will be found so much in line with observations which all who have used these agents must have made—although, perhaps, they may not have paused to analyze them—that their truth and value will be readily understood, and may help in guiding aright others with less experience in this line.

It has been truly said that it is as wrong as it is useless to attempt to convince through argument, but that one's duty is to offer evidence and let the unprejudiced convince themselves. This principle applies very strongly to those who, despite what one would think are insuperable objections, make use of stock vaccines. No one for a moment would imagine that an attack of measles, although known to confer great specific immunity, could aid in preventing or in overcoming a typhoid infection; and yet if we employ a stock vaccine, in at least ninety-nine instances out of a hundred we follow a similar procedure. Let any one who has fixed on a variety of stock vaccine for use in a given case look up the possible bacteriology of the disease process which he proposes to treat, and then calculate his chances of striking, even by name, the guilty organism, let alone the particular strain active—which may even be one as yet unrecognized by science—the last consideration a factor which so often confuses exact bacteriologic diagnosis and serum therapy, but which is entirely avoided in vaccine treatment when autogenous products are used.

The fact that a given stock is composed of a number of strains cannot reassure, since at least all but one would be given needlessly and could not but burden, if they did not seriously injure, the patient's reactive mechanism—as in the case of the rabbit dosed with colon bacilli in order to render him susceptible to organisms to which he is normally immune, an effect which could be produced by the use of any stock vaccine on the market. Besides, the polyvalence of most vaccines relies solely on difference in source and not on actual known biologic differences